

Appendix A. Notations and Definitions

Hyperspace: The hyperspace $\mathcal{H}^D$ consists of all bipolar hypervectors of dimension D , where each entry takes a value from the set $\{+1, -1\}$. Formally, we define:

$$\mathcal{H}^D := \{-1, +1\}^D = \{A \in \mathbb{R}^D \mid a_i \in \{+1, -1\}, \forall i = 1, \dots, D\} \quad (\text{A.1})$$

with a_i representing the i -th component of hypervector A .

Unless stated otherwise, for a *random hypervector* $A \in \mathcal{H}^D$, each component a_i is independently sampled from a symmetric binary distribution:

$$\mathbb{P}(a_i = +1) = \mathbb{P}(a_i = -1) = 0.5, \quad i = 1, \dots, D. \quad (\text{A.2})$$

Elementary Functions: For any hypervectors $A, B \in \mathcal{H}^D$, we introduce the following basic operations:

- Count of $+1$ entries in A :

$$\text{pos}(A) := \sum_{i=1}^D \mathbb{I}(a_i = +1) \quad (\text{A.3})$$

- Count of -1 entries in A :

$$\text{neg}(A) := \sum_{i=1}^D \mathbb{I}(a_i = -1) \quad (\text{A.4})$$

- Binding (element-wise multiplication) of A and B :

$$A \otimes B := A \times B \quad (\text{A.5})$$

- Bundling (element-wise majority) of A and B :

$$A \oplus B := [A + B] \quad (\text{A.6})$$

Here, $\mathbb{I}(\cdot)$ is the indicator function, $\times$ denotes element-wise multiplication, $+$ is element-wise addition, and $[\cdot]$ represents the element-wise majority function, which outputs $+1$ if the sum is positive, -1 if negative, and randomly selects between $\{-1, +1\}$ when the sum is zero.

Hamming Distance: For two hypervectors $A, B \in \mathcal{H}^D$, the Hamming distance measures the fraction of positions where they differ:

$$d_H : \mathcal{H}^D \times \mathcal{H}^D \rightarrow [0, 1], \quad (\text{A.7})$$

$$d_H(A, B) = \frac{1}{D} \sum_{i=1}^D \mathbb{I}(a_i \neq b_i) = \frac{1}{D} \text{neg}(A \otimes B). \quad (\text{A.8})$$

Level Set: A set of M hypervectors $\mathcal{L} = \{L^{(j)}\}_{j=1}^M \subset \mathcal{H}^D$ is called a level set if it is constructed as follows:

- Sample a base hypervector $L^{(1)} \in \mathcal{H}^D$ randomly.
- Initialize a set $\mathcal{B} = \emptyset$ to track the flipped positions.
- For $i = 2, \dots, M$:
 - Randomly select $\frac{D}{M-1}$ positions from $\{1, \dots, D\} \setminus \mathcal{B}$ (positions not yet flipped). Assume $\frac{D}{M-1} \in \mathbb{N}$ for simplicity.
 - Flip the selected bits in $L^{(i-1)}$ to generate $L^{(i)}$.
 - Update $\mathcal{B}$ to include the newly flipped positions.

Under this construction, the Hamming distance between any two hypervectors $L^{(i)}$ and $L^{(j)}$ is

$$d_H(L^{(i)}, L^{(j)}) = \frac{|i - j|}{M - 1}. \quad (\text{A.9})$$

Mapping with Levels: Let $\mathcal{X} \subseteq \mathbb{R}^d$ denote the feature space. Consider d random hypervectors $\mathcal{ID} = \{ID^{(1)}, \dots, ID^{(d)}\} \subset \mathcal{H}^D$ and a level set $\mathcal{L} = \{L^{(1)}, \dots, L^{(M)}\} \subset \mathcal{H}^D$.

For each feature dimension $n = 1, \dots, d$, define thresholds $\theta_{n,1} < \dots < \theta_{n,M-1}$ to partition $\mathbb{R}$ into M intervals:

$$I_0 = (-\infty, \theta_{n,1}), \quad I_1 = [\theta_{n,1}, \theta_{n,2}), \dots, I_{M-1} = [\theta_{n,M-1}, +\infty) \quad (\text{A.10})$$

Given a feature value $x_i[n]$, assign it to the unique interval I_m such that $x_i[n] \in I_m$, defining

$$l_n(x) = \begin{cases} 1, & x < \theta_{n,\alpha} \\ \lfloor \frac{x - \theta_{n,\alpha}}{\theta_{n,\beta} - \theta_{n,\alpha}} \cdot M + 1 \rfloor, & x \in [\theta_{n,\alpha}, \theta_{n,\beta}) \\ M, & x \geq \theta_{n,\beta} \end{cases}, \quad (\text{A.11})$$

where $\theta_{n,\alpha}$ and $\theta_{n,\beta}$ are the 2% and 98% quantiles of feature n .

Appendix B. Lemmas and Propositions

Lemma 1 (Hamming Distance between product and multiplier).

For hypervectors $A, B \in \mathcal{H}^D$, the Hamming distance between A and $A \times B$ depends only on B :

$$d_H(A, A \times B) = \frac{1}{D} \text{neg}(B). \quad (\text{B.1})$$

Proof. By the definition of element-wise multiplication, the i -th component satisfies

$$a_i b_i = \begin{cases} a_i, & b_i = +1, \\ -a_i, & b_i = -1, \end{cases} \quad (\text{B.2})$$

and equivalently,

$$a_i \neq a_i b_i \iff b_i = -1. \quad (\text{B.3})$$

Thus, the Hamming distance can be written as

$$d_H(A, A \times B) = \frac{1}{D} \sum_{i=1}^D \mathbb{I}(a_i \neq a_i b_i) = \frac{1}{D} \sum_{i=1}^D \mathbb{I}(b_i = -1) = \frac{1}{D} \text{neg}(B). \quad (\text{B.4})$$

Lemma 2 (Hamming Distance Preservation under Multiplication). For hypervectors $A, B, C \in \mathcal{H}^D$:

$$d_H(A \times B, B \times C) = d_H(A, C). \quad (\text{B.5})$$

Proof. Considering component-wise multiplication:

$$a_i b_i \neq b_i c_i \iff a_i \neq c_i, \quad (\text{B.6})$$

and therefore

$$d_H(A \times B, B \times C) = \frac{1}{D} \sum_{i=1}^D \mathbb{I}(a_i b_i \neq b_i c_i) = \frac{1}{D} \sum_{i=1}^D \mathbb{I}(a_i \neq c_i) = d_H(A, C). \quad (\text{B.7})$$

Lemma 3 (Hamming distance between two random hypervectors). Let $A, B \in \mathcal{H}^D$ be independent random hypervectors. Then, for any $\epsilon > 0$,

$$\lim_{D \rightarrow \infty} \mathbb{P}(|d_H(A, B) - 0.5| > \epsilon) = 0. \quad (\text{B.8})$$

Proof. Define indicator variables $X_i := \mathbb{I}(a_i \neq b_i)$, so that

$$d_H(A, B) = \frac{1}{D} \sum_{i=1}^D X_i. \quad (\text{B.9})$$

Since each a_i and b_i is i.i.d. from $\{-1, +1\}$, each X_i is an independent Bernoulli variable with mean 0.5. By the Weak Law of Large Numbers:

$$\lim_{D \rightarrow \infty} \mathbb{P}\left(\left|\frac{1}{D} \sum_{i=1}^D X_i - 0.5\right| > \epsilon\right) = 0, \quad (\text{B.10})$$

so the Hamming distance concentrates around 0.5.

Lemma 4 (Hamming distance between a random hypervector and its product with another). Let $A, B \in \mathcal{H}^D$ be independent random hypervectors. Then, for any $\epsilon > 0$,

$$\lim_{D \rightarrow \infty} \mathbb{P}(|d_H(A, A \times B) - 0.5| > \epsilon) = 0. \quad (\text{B.11})$$

Proof. From Lemma 1, we have

$$d_H(A, A \times B) = \frac{\text{neg}(B)}{D}. \quad (\text{B.12})$$

Since B is random with each component i.i.d., applying the Weak Law of Large Numbers gives

$$\lim_{D \rightarrow \infty} \mathbb{P}\left(\left|\frac{\text{neg}(B)}{D} - 0.5\right| > \epsilon\right) = 0, \quad (\text{B.13})$$

which implies the result.

Proposition (Statistical view of Hamming Distance). Consider hypervectors $A, B \in \mathcal{H}^D$ with

$$d_H(A, B) = \delta, \quad (\text{B.14})$$

and define

$$Z_i = \mathbb{I}(a_i \neq b_i) \quad (\text{B.15})$$

to indicate whether the i -th bit differs. For large D , we can approximate

$$Z_i \sim_{i.i.d.} \text{Bernoulli}(\delta), \quad i = 1, \dots, D. \quad (\text{B.16})$$

Proof. By definition,

$$\frac{1}{D} \sum_{i=1}^D Z_i = d_H(A, B) = \delta. \quad (\text{B.17})$$

Although the Z_i are not strictly independent (their sum is fixed), for large D :

- The probability that the empirical mean deviates from δ is negligible:

$$\mathbb{P}\left(\left|\frac{1}{D} \sum_{i=1}^D Z_i - \delta\right| > \epsilon\right) \leq 2 \exp(-2D\epsilon^2). \quad (\text{B.18})$$

- If A and B are generated randomly conditioned on their Hamming distance, the mismatch positions are uniformly random.

Hence, Z_i can be treated as i.i.d. Bernoulli variables for large D .